

22-2 풀이시험

7월 1주

1. The joint probability density function (결합확률밀도함수) of random variables X and Y is given as

$$f(x, y) = \begin{cases} x + y, & 0 \leq x \leq 1, 0 \leq y \leq 1; \\ 0, & \text{elsewhere;} \end{cases}$$

Answer the following questions.

- (1) Marginal probability density function (여백밀도함수) of X is given as

$$f(x) = \begin{cases} ax + b, & 0 \leq x \leq 1; \\ 0, & \text{elsewhere;} \end{cases}$$

Find a and b .

- (2) Calculate $E(x)$ and $Var(X)$.

- (3) Conditioned on $X=x$, calculate the conditional probability density function (확률밀도함수) $f(y|x)$.

$$1. (1) f(x) = \int_{-\infty}^{\infty} f(x, y) dy = \int_0^1 (x+y) dy = \left[xy + \frac{1}{2}y^2 \right]_0^1 = x + \frac{1}{2}$$

$$a=1, b=\frac{1}{2}$$

$$(2) E(X) = \int_0^1 x \left(x + \frac{1}{2}\right) dx = \left[\frac{1}{3}x^3 + \frac{1}{2}x^2 \right]_0^1 = \frac{1}{3} + \frac{1}{2} = \frac{5}{6}$$

$$E(X^2) = \int_0^1 x^2 \left(x + \frac{1}{2}\right) dx = \left[\frac{1}{4}x^4 + \frac{1}{6}x^3 \right]_0^1 = \frac{1}{4} + \frac{1}{6} = \frac{5}{12}$$

$$V(X) = E(X^2) - (E(X))^2 = \frac{5}{12} - \left(\frac{5}{6}\right)^2 = \frac{11}{144}$$

$$(3) 0 \leq x \leq 1, 0 \leq y \leq 1 \text{ on } (1), f(y|x) = \frac{f(x, y)}{f(x)} = \frac{x+y}{x+\frac{1}{2}} = \frac{2(x+y)}{2x+1}$$

$$\therefore f(y|x) = \begin{cases} \frac{2x+2y}{2x+1}, & 0 \leq y \leq 1 \\ 0, & \text{o.w.} \end{cases}$$

2. Assume that the word 'free' occurs in 70% of the spam e-mail. We also assume that the word 'free' occurs in 20% of the non-spam e-mails. It is known that 10% of the received e-mails are considered as a spam e-mail. Answer the following questions.

- (1) What is a probability that the randomly chosen email has a word 'free'?
- (2) Assuming that a new e-mail containing a word 'free', what is the probability that it is spam e-mail ?

$$2. \Delta \text{스팸이메일일 확률: } P(S) = 0.1$$

$$\Delta \text{스팸이 아닐(정상) 확률: } P(S^c) = 1 - 0.1 = 0.9$$

$$\Delta \text{스팸이메일 중 'free'가 포함될 확률: } P(F|S) = 0.7$$

$$\Delta \text{스팸이 아닐 이메일 중 'free'가 포함될 확률: } P(F|S^c) = 0.2$$

$$(1) P(F) = P(F|S)P(S) + P(F|S^c)P(S^c) = (0.7 \times 0.1) + (0.2 \times 0.9) = 0.25 \quad (\because \text{Law of Total Probability})$$

$$(2) P(S|F) = \frac{P(F|S)P(S)}{P(F)} = \frac{0.7 \times 0.1}{0.25} = \frac{0.07}{0.25} = 0.28 \quad (\because \text{Bayes' Theorem})$$

3. Consider an Exponential random variable X of mean $1/3$.

(1) Calculate the probability density function of X at $X = x$.

(2) Calculate the moment generating function $M_X(t) = E(e^{tX})$ for $3 - t > 0$.

(3) Using $M_X(t)$ calculated in the above, calculate $E(X^2)$.

$$3. (1) f(x) = \begin{cases} 3e^{-3x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

$$(2) M_X(t) = E(e^{tx}) = \int_0^{\infty} e^{tx} \cdot 3e^{-3x} dx = 3 \int_0^{\infty} e^{-(3-t)x} dx = 3 \left[\frac{-1}{3-t} e^{-(3-t)x} \right]_0^{\infty} = 3 \left(0 - \left(-\frac{1}{3-t} \right) \right) = \frac{3}{3-t}$$

$$M_X(t) = \frac{3}{3-t} \quad (t < 3)$$

$$(3) E(X^2) = M_X''(0)$$

$$M_X(t) = 3(3-t)^{-1}$$

$$M_X'(t) = -3(3-t)^{-2} = -3(3-t)^{-2}$$

$$M_X''(t) = -6(3-t)^{-3}(-1) = 6(3-t)^{-3}$$

$$E(X^2) = 6(3-0)^{-3} = \frac{6}{27} = \frac{2}{9}$$

수리통계학 II 및 연습

2) Let $X_1, \dots, X_n$ be a random sample from the **exponential** distribution with the p.d.f.

$$f(x|\theta) = \theta e^{-\theta x}, \quad x > 0, \quad \theta > 0.$$

a) Find the **likelihood function** $L(\theta|x_1, \dots, x_n)$. (hint : $L(\theta|x_1, \dots, x_n) = f(x_1|\theta) \cdots f(x_n|\theta)$)

$$a) L(\theta|x_1, \dots, x_n) = \prod_{i=1}^n f(x_i|\theta) = \prod_{i=1}^n \theta e^{-\theta x_i} = \theta^n e^{-\theta \sum_{i=1}^n x_i}$$

b) Find the **MLE**(**maximum likelihood estimator**) $\hat{\theta}_n$ of θ .

$$(\text{hint : } \partial \ln L(\theta|x_1, \dots, x_n) / \partial \theta = \sum_{i=1}^n \partial [\ln f(x_i|\theta)] / \partial \theta = 0)$$

$$b) \ln L(\theta|x_1, \dots, x_n) = \ln(\theta^n e^{-\theta \sum_{i=1}^n x_i}) = n \ln \theta - \theta \sum_{i=1}^n x_i$$

$$\frac{\partial}{\partial \theta} \ln L(\theta|x_1, \dots, x_n) = \frac{n}{\theta} - \sum_{i=1}^n x_i = 0$$

$$\frac{n}{\theta} = \sum_{i=1}^n x_i \Rightarrow \hat{\theta}_n = \frac{n}{\sum_{i=1}^n x_i} = \frac{1}{\bar{x}}$$

c) Find the approximate **95% confidence interval** for θ for large n.

(hint: $\hat{\theta}_n \sim N(\theta, [nI_1(\theta)]^{-1})$ for large n ; i.e. $P(-1.96 < Z_n < 1.96) = .95$, $Z_n = [\hat{\theta}_n - \theta] / [1/n I_1(\hat{\theta}_n)]^{1/2} \sim N(0, 1)$)

where $I_1(\theta) = E[-\partial^2 \ln f(X_1|\theta) / \partial \theta^2 | \theta]$, $E(X_1|\theta) = 1/\theta$, $\text{Var}(X_1|\theta) = 1/\theta^2$)

$$c) \ln f(X_1|\theta) = \ln \theta - \theta X_1$$

$$\frac{\partial}{\partial \theta} \ln f(X_1|\theta) = \frac{1}{\theta} - X_1$$

$$\frac{\partial^2}{\partial \theta^2} \ln f(X_1|\theta) = -\frac{1}{\theta^2}$$

$$I_1(\theta) = E\left[-\frac{\partial^2}{\partial \theta^2} \ln f(X_1|\theta)\right] = E\left[\frac{1}{\theta^2}\right] = \frac{1}{\theta^2}$$

$$\hat{\theta}_n \sim N\left(\theta, \frac{1}{n I_1(\theta)}\right)$$

$$P(-1.96 < \frac{\hat{\theta}_n - \theta}{\sqrt{\frac{1}{n I_1(\hat{\theta}_n)}}} < 1.96) \approx 0.95$$

$$95\% \text{ CI} = \left(\hat{\theta}_n - 1.96 \frac{\hat{\theta}_n}{\sqrt{n}}, \hat{\theta}_n + 1.96 \frac{\hat{\theta}_n}{\sqrt{n}}\right)$$